Computational Sufficiency of Recurrent Perturbations for Selected Acute Psilocybin Working-Memory Signatures

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Abstract deferred until the end of the project. Draft only after the specificity and dynamics sections have evidence.

CONTENTS

INTRODUCTION

Working memory requires a neural system to preserve task-relevant information after its eliciting input has disappeared. Recurrent excitation, attractor dynamics, and slow manifolds provide candidate mechanisms for that maintenance [1, 2]. A weak change in those dynamics need not erase memory at once; it may instead expose selective vulnerability when information must be held longer, protected from irrelevant input, or updated under greater load.

Acute human psilocybin studies motivate exactly such selective targets rather than global cognitive collapse: clearer impairment under higher updating load, greater vulnerability when competing information must be ignored, stronger effects at longer intervals or higher dose, and response slowing that can outpace accuracy loss [3–7]. Psychedelic theory and modelling further motivate competing computational interventions—especially gain-like and weighting changes—without specifying a unique working-memory RNN equation [8, 9]. Section reviews that evidence in full.

This dissertation therefore uses task-trained recurrent networks as a controlled testbed. The aim is computational sufficiency and comparative mechanism: which literature-motivated perturbations, if any, reproduce the selected behavioural pattern more convincingly than generic disruption, and how that pattern emerges from recurrent population dynamics (Section). Methods, the completed candidate screen, and the outstanding specificity and dynamics experiments are developed in Sections –.

LITERATURE REVIEW

This section synthesises the core literature that motivates the present computational study. The review is organised around project logic rather than paper-by-paper summary: working memory as recurrent dynamics; trained RNNs as testbeds; acute psilocybin behavioural signatures; psychedelic neural-dynamics theories; and serotonergic gain mechanisms. Direct evidence, modelling analogies, and speculative translations are kept distinct.

Working memory as recurrent dynamics

Working memory can be supported by recurrent excitation that sustains task-relevant activity after sensory drive has disappeared. In prefrontal cortex, NMDA-dependent recurrent mechanisms contribute to delay-period persistent firing [1]. At the population level, however, stable coding need not require every neuron to fire tonically: Murray and colleagues showed that heterogeneous single-unit trajectories can coexist with a low-dimensional subspace in which stimulus representations remain decodable across cue and delay epochs [10]. Theoretical work likewise emphasises that recurrent heterogeneity and attractor structure shape drift, noise robustness, and memory stability [11].

Trained recurrent networks provide a complementary modelling route. Song, Yang, and Wang showed that excitatory–inhibitory RNNs can be trained on cognitive tasks, including parametric working memory, while remaining useful as mechanistic hypothesis generators provided that dynamics and connectivity are analysed rather than behaviour alone [12]. Ghazizadeh and Ching further showed that trained working-memory networks can encode information along slow manifolds, with multiple dynamical solutions capable of supporting delay-period performance and different robustness/forgetting tradeoffs [2]. Frontoparietal circuit models also motivate testing maintenance under distraction and distributed recurrent roles [13]. Together, these sources justify a task-trained RNN baseline judged by both behavioural competence and population-level delay dynamics.

Acute psilocybin effects on attention and working memory

Human psilocybin cognition does not reduce to a single global impairment. Across attention and executive tasks, a multilevel meta-analysis found a large acute slowing of reaction time (Hedges' $g = 1.13$, 95% CI $[0.57, 1.70]$) with a weaker and less robust accuracy effect [7]. Dose dependence and task sensitivity were important: broader or more demanding measures tended to show clearer slowing. This is the strongest behavioural rationale for including timing-sensitive outcomes, such as response settling, alongside final accuracy.

Original studies refine the selective pattern. In a letter N-back task, psilocybin impaired updating more clearly at 2-back than at 0-back, with reduced discriminability and prolonged responding [3]. In a small within-subject study, multiple-object tracking was impaired while spatial span was comparatively spared [4]. Interval reproduction and spatial span showed stronger impairment at higher dose and longer intervals [5]. On a memory-guided delayed-response task, reaction time increased without a clear reduction in correct responses [6]. Broader reviews likewise report mixed working-memory findings and more consistent acute attentional vulnerability, while cautioning against equating acute healthy-volunteer effects with post-acute clinical outcomes [14, 15]. Sayali and Barrett additionally frame psychedelic cognition as a stability–flexibility tradeoff rather than uniform deficit [15].

For modelling, these findings support a selective target: greater cost under updating load and distraction, response slowing with relative preservation of simple accuracy, and stronger impairment at greater perturbation strength. They do not specify an RNN equation.

Psychedelic neural-dynamics theories and models

Theoretical accounts motivate altered precision, gain, and constraint. The entropic-brain framework emphasised increased signal diversity and reduced ordered constraint under psychedelics [16]. REBUS relates relaxed high-level prior precision to altered postsynaptic gain and greater influence of afferent prediction-error signals [8]. These are interpretive frameworks, not working-memory implementations.

Computational and empirical models sharpen the translation space. Herzog and colleagues operationalised 5-HT2A effects in a whole-brain dynamic mean-field model as receptor-density-dependent modulation of the neural response function, producing heterogeneous entropy increases shaped by recurrent topology [9]. Empirical MEG work similarly reports increased spontaneous signal diversity under psychedelic doses of several compounds, including psilocybin [17]. Stoliker et al. analysed psilocybin fMRI/EEG across rest, meditation, music, and movie contexts and argued that variability can remain structured and context-sensitive rather than randomly disorganised; stronger subjective effects accompanied more cohesive context-specific trajectories [18]. Bredenberg et al. modelled a shift toward top-down generative influence and showed that structured internal weighting can produce variability and distortions distinct from additive noise controls [19].

None of these sources is a working-memory RNN. Their joint implication is nonetheless clear: candidate perturbations should include gain-like and weighting manipulations, and generic noise should be retained as a control rather than treated as the default psychedelic mechanism.

Serotonergic and gain mechanisms relevant to working memory

Cellular and circuit evidence makes response gain a particularly tractable bridge. In rat prefrontal pyramidal neurons, serotonin increased the slope of the firing-rate/current relationship through postsynaptic 5-HT2 mechanisms [20]. In monkeys performing oculomotor delayed response, local 5-HT2A stimulation and blockade altered delay-period spatial tuning in a dose-dependent and non-monotonic manner involving both putative excitatory and inhibitory cells [21]. Cano-Colino and colleagues used a receptor-informed spatial working-memory attractor model to show that serotonergic manipulations can produce distinct failure modes—delay decay, spurious bumps, distractibility, or perseveration—and that isolated 5-HT2A agonism increased distractor resistance in their model rather than simple distractor capture [22]. That result is an important constraint: behavioural distractor vulnerability under psilocybin should not be equated a priori with a receptor claim that 5-HT2A agonism increases distractor gain.

Direct psilocin work strengthens heterogeneity. Schmitz et al. found that psilocin excited identified medial-prefrontal 5-HT2A neurons while producing mixed increases, decreases, and null effects across the broader pyramidal population [23]. Human cortical-propagation analyses further suggest that psilocybin effects depend on the heterogeneous 5-HT2A receptor landscape [24]. These findings motivate testing both global and unit-wise heterogeneous drive-gain analogues, while treating sensory, distractor, and recurrent weightings as separate competing translations.

Evidence gaps and modelling implications

Across this core set, no reviewed source both trains a working-memory RNN and tests which literature-motivated dynamical perturbation recovers selective acute psilocybin behavioural signatures more convincingly than generic disruption. To our knowledge, that combined question remains open. The literature therefore justifies the present programme as computational sufficiency and comparative mechanism, not as biological identification: build competent delay-period networks; define selective behavioural contrasts from human task evidence; compare gain, afferent, recurrent, persistence, and timescale interventions against a native baseline and, later, against matched Gaussian disruption; and explain any selective match through population dynamics rather than accuracy alone.

AIMS, RESEARCH QUESTIONS, AND CLAIM BOUNDARY

Adopted end goal

The end goal of this dissertation is to determine whether a literature-motivated change in network dynamics can reproduce a selective pattern of acute psilocybin-related working-memory effects more convincingly than generic neural disruption, and to explain how that pattern emerges from the RNN's internal dynamics. The project is not intended as a literal pharmacological simulation of psilocybin. A task-trained RNN is used as a controlled computational testbed for comparing candidate mechanisms.

Required contributions

1. Establish a credible working-memory baseline across independently trained checkpoints.
2. Define a nonspecific perturbation control (independent Gaussian noise), retaining structured-noise work as background robustness evidence rather than as a direct psilocybin model.
3. Compare literature-grounded candidate perturbations against selective behavioural signatures: distractor vulnerability, high-load updating cost, response slowing or longer settling, relative preservation under weaker perturbation, and stronger impairment at greater strengths.
4. Provide a dynamical explanation in terms of delay-period decoding, distractor displacement and recovery, representational drift, memory-subspace geometry, and related hidden-state diagnostics.

Working research questions

1. Which single-operator RNN perturbations, if any, reproduce the selected cross-task behavioural pattern

against the native baseline?

2. Does the leading candidate exceed matched-cost Gaussian disruption?

3. What changes in recurrent population dynamics explain any selective behavioural match?

Provisional claim map

Table I records the present evidential status. Claims that depend on unfinished experiments remain open.

Inferential boundaries

The dissertation should not claim that:

- the RNN is a biological simulation of psilocybin;
- one perturbation strength corresponds to a human dose;
- behavioural similarity identifies the true neural mechanism;
- increased entropy alone makes a model psychedelic-like;
- a null specificity result is experimental failure rather than a scientifically informative constraint.

COMPUTATIONAL METHODS

Fixation-gated circular delayed-response task

Each circular-task trial comprised pre-cue fixation, a circular cue, a silent memory delay, and a response period (Fig. 1A). Stimulus angle $\theta \in [0, 2\pi)$ was encoded over 32 evenly spaced preferred angles μ_j as

$$c_j(\theta) = \exp\{\kappa[\cos(\theta - \mu_j) - 1]\}, \qquad \kappa = 8. \qquad (1)$$

The fixation input and output remained high until the response cue. Circular outputs were deliberately silent before response and then reported the remembered angle. Pre-cue and cue durations varied during training; evaluation used clean delays of 10, 20, 40, and 80 model steps. A separate delay-20 condition inserted a five-step irrelevant circular cue midway through the delay. Training sampled one clean and one distractor-present homogeneous batch exactly once per shuffled two-batch block. Distractor filtering was therefore a learned property of the evaluated networks rather than an out-of-distribution challenge.

Context-cued N-back task

The N-back task presented 20 sequential items drawn from six stimulus identities (Fig. 1B). Two context channels instructed either 0-back or 2-back behaviour. In 0-back, the network classified whether the current item matched a fixed target identity. In 2-back, it classified whether the current item matched the item presented two positions previously. Sequences contained six matches; the 2-back generator additionally controlled one-back lures. The first two items were excluded from scored outcomes. The output consisted of match and non-match logits.

Network architecture

Both task families used the same rate-style continuous-time recurrent network (CTRNN) family: a leaky recurrent hidden layer with a linear readout. The model is not spiking, not Dale-constrained, and not intended as a biophysically detailed cortical circuit. It is a controlled dynamical testbed with an explicit time constant so that maintenance stability can be varied computationally.

Let $x_t \in \mathbb{R}^{N_{\text{in}}}$ be the input at discrete time t, $h_t \in \mathbb{R}^{N_h}$ the hidden state, and $y_t \in \mathbb{R}^{N_{\text{out}}}$ the readout. The native recurrence mixes afferent and recurrent drive,

$$z_t = W_{\text{in}} x_t + b_{\text{in}} + W_{\text{rec}} h_{t-1} + b_{\text{rec}}, \qquad (2)$$

then applies a leaky update with nonlinearity after mixing,

$$h_t = \tanh[(1-\alpha)h_{t-1} + \alpha z_t], \qquad \alpha = \frac{dt}{\tau}. \qquad (3)$$

Here $dt = 20$ and $\tau = 100$, so $\alpha = 0.2$. The factor $(1-\alpha)$ is the carried-state weight and α the drive weight. Because tanh acts after leak mixing rather than only on z_t, later gain and persistence operators are translations of this implemented update, not exact maps of a Song-style $\tau\dot{h} = -h + f(z)$ form. Hidden states were initialised at zero at the start of each trial. Outputs were a linear readout of the hidden state,

$$y_t = W_{\text{out}} h_t + b_{\text{out}}. \qquad (4)$$

Unless otherwise stated, $N_h = 64$ and the hidden nonlinearity was tanh. The circular model used $N_{\text{in}} = N_{\text{out}} = 33$ (32 tuned channels plus fixation). The N-back model used eight inputs (six stimulus identities and two rule-context channels) and two outputs (non-match and match logits). Learned weights were frozen before all perturbation analyses.

Training and checkpoint replication

Throughout, a *checkpoint* means one independently trained network with frozen weights, identified by its ran-

Table I. Provisional claim map. Status distinguishes what the present candidate screen supports from what still requires Sections and .

Claim	Present status	Main support / blocker
Competent circular and N-back baselines exist	Supported	Section ; retained checkpoints
Persistence 0.95 meets the descriptive majority pattern vs native baseline	Supported, provisional	Section ; weak circular intervals
N-back load selectivity under persistence 0.95 is robust	Supported	10/10 seeds; interval excludes zero
Circular settling/delay/distractor effects are robust	Not supported yet	Seed-level intervals include zero
Circular and N-back hidden-state geometry is interpretable at baseline	Supported, descriptive	Section ; representative seeds
Persistence exceeds matched-cost Gaussian disruption	Open	Section not run
Selective behaviour is explained by recurrent dynamics	Open	Section not run
Biological identification of psilocybin mechanism	Out of scope	Claim boundary below

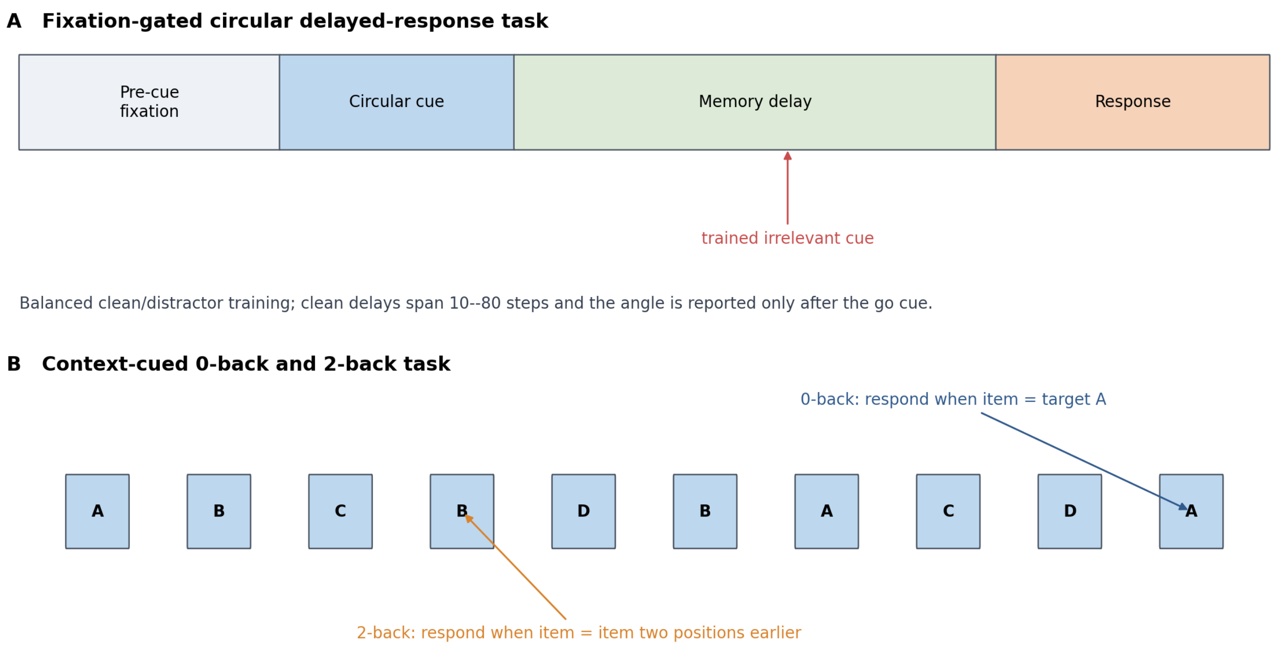

Figure 1. Complementary working-memory tasks. **A**, The fixation-gated circular task required a continuous cue angle to be maintained across variable delays and reported after the go cue; balanced clean/distractor training made resistance to an irrelevant delay-period cue a learned behaviour. **B**, Context inputs switched the same N-back network between a fixed-target 0-back rule and a 2-back updating rule. The tasks supplied settling, delay, distractor, and updating-load contrasts rather than a single performance level.

dom seed. Checkpoint, not trial or technical replicate, is the replication unit: summary statistics are formed across trained seeds, and within-seed trial averages are treated as measurement precision rather than independent scientific replicates.

Circular networks were trained for 4,000 Adam updates (learning rate 10^{-3}, batch size 64) with Yang-style weighted fixation and response loss on a balanced clean/distractor mixture; recurrent training noise standard deviation was 0.05. Six circular seeds were trained; five met preconfigured competence limits (clean mean error $\leq 10°$, distractor mean error $\leq 15°$, fixation accuracy ≥ 0.90) and were retained (seeds 20260731, 20260732, 20260733, 20260735, and 20260736). Seed 20260734 exceeded those limits (clean 15.75°, distractor 17.46°) and was excluded before the perturbation screen. The ten N-back checkpoints (seeds 20260912–20260921) were drawn from a competence-screened pool trained with staged 0-back then 2-back curriculum under Adam (learning rate 10^{-3}, no recurrent training noise). Final N-back retention required accuracy ≥ 0.95, discriminability ≥ 0.90, and 2-back lure accuracy ≥ 0.90; stage-1 0-back acquisition used stricter gates (accuracy ≥ 0.98, discriminability ≥ 0.95).

These gates are operational competence thresholds, not fitted human effect sizes. Chance circular error is of order 90°, so clean $\leq 10^\circ$ requires unambiguous angle memory while remaining softer than the few-degree errors typical of retained seeds; distractor $\leq 15^\circ$ allows a modest filtering cost without accepting near-failure. Fixation ≥ 0.90 keeps the network in a reliable go regime so latency contrasts are not confounded by fixation collapse. N-back gates demand near-ceiling fixed-target performance, clearly above-chance updating, and lure resistance, so load contrasts reflect selective updating cost rather than a weak baseline.

Prior baseline and structured-noise background

Earlier project stages established that continuous circular delayed response is learnable in the same CTRNN family and that noise structure matters for representational robustness. A categorical tanh delay baseline reached perfect held-out accuracy across tested delays, while a tuned circular baseline supported stable angle decoding with delay-period slowing of hidden-state speed. Those reports remain methodological background rather than the psilocybin-signature screen.

A five-seed structured-noise study on Yang-style fixation-gated circular checkpoints compared matched-RMS independent Gaussian, temporal AR(1), and topology-correlated AR(1) hidden-state noise. At equal energy, impairment followed independent < temporal < topology-correlated structure, grew with delay, accumulated during maintenance, and coincided with collapse of the circular memory geometry. That hierarchy is retained as a robustness result and as motivation for treating unstructured Gaussian noise as the generic control in Section . Temporally persistent and topology-aligned noise are not used as primary psychedelic candidates in the present signature screen: they destroy representations efficiently, but they are not the literature-preferred mechanism class for selective behavioural dissociations.

Candidate perturbations

We evaluated seven single-operator families against each checkpoint's native baseline (Table II). Operators are competing computational interventions motivated by, but not equated with, psychedelic or working-memory mechanisms. Because the native update applies tanh after mixing drive with carried state, no operator is an exact biological response-gain map; each is an explicit RNN translation. Unless noted, every non-neutral operator was applied at every timestep of the full frozen forward pass for the entire trial or sequence; learned weights were never updated. The only timed exception is P3b, which is confined to the irrelevant circular-cue window. Write $z_t^{\mathrm{w}} = W_{\mathrm{in}}x_t + W_{\mathrm{rec}}h_{t-1}$ for the weight-derived drive. Symbols are shared across perturbations: x_t is input, h_t hidden state, and z_t pre-activation drive at time t; W_{in} and W_{rec} are input and recurrent weights; b_{in} and b_{rec} are biases; and $\alpha = dt/\tau$ is the native leak factor. In P3a/P3b, x_t^{stim} and x_t^{ctrl} denote stimulus versus fixation/rule-control channels, with matching input submatrices $W_{\mathrm{in}}^{\mathrm{stim}}$ and $W_{\mathrm{in}}^{\mathrm{ctrl}}$. Scalar g denotes uniform gain, vector $\mathbf{g}$ fixed unit-wise gain (P2), $\varepsilon_t \sim \mathcal{N}(0, I)$ Gaussian noise scaled by σ (P5), p persistence scale (P6), and s timescale factor with derived α' (P7). Unless noted, learned biases stay outside multiplicative gains.

P1 Synaptic-drive gain

Herzog et al. operationalised 5-HT2A effects as response gain on synaptic current inside a nonlinearity [9]; REBUS likewise links altered precision to postsynaptic gain [8]. In this CTRNN the closest available analogue multiplies weight-derived drive while leaving biases unscaled:

$$h_t = \tanh\left[(1-\alpha)h_{t-1} + \alpha\left(g z_t^{\mathrm{w}} + b_{\mathrm{in}} + b_{\mathrm{rec}}\right)\right]. \quad (5)$$

This is synaptic-drive gain, not an exact F–I slope change and not a pure carried-state manipulation.

P2 Heterogeneous drive gain

Receptor-density dependence in Herzog et al. implies spatially nonuniform gain rather than one global scalar [9]. We therefore replaced g by a fixed positive unit-wise vector $\mathbf{g}$ drawn from a log-normal distribution and normalised to population mean one:

$$h_t = \tanh\left[(1-\alpha)h_{t-1} + \alpha\left(\mathbf{g} \odot z_t^{\mathrm{w}} + b_{\mathrm{in}} + b_{\mathrm{rec}}\right)\right]. \quad (6)$$

Three fixed vectors were treated as technical replicates and averaged within checkpoint; they are not independent trained seeds.

P3a Sensory-input gain

REBUS predicts increased influence of afferent or bottom-up signals when high-level precision is relaxed [8]. As a task-level translation, only stimulus-channel input weights were scaled, leaving fixation or rule-context channels unchanged:

$$\begin{aligned} z_t^{\mathrm{in}} &= g\, W_{\mathrm{in}}^{\mathrm{stim}} x_t^{\mathrm{stim}} + W_{\mathrm{in}}^{\mathrm{ctrl}} x_t^{\mathrm{ctrl}} + b_{\mathrm{in}}, \\ h_t &= \tanh\left[(1-\alpha)h_{t-1} + \alpha(z_t^{\mathrm{in}} + W_{\mathrm{rec}}h_{t-1} + b_{\mathrm{rec}})\right]. \end{aligned} \quad (7)$$

This tests sensory sensitivity, not receptor-mapped response gain.

P3b Distractor-window gain

Human psilocybin work shows particular vulnerability when competing information must be ignored [4]. P3b applies Eq. (7) only during the irrelevant circular cue and is therefore a circular-task manipulation check rather than a complete cross-task mechanism candidate.

P4 Recurrent gain

Recurrent excitation is a canonical substrate of delay-period maintenance [1]. To separate recurrent weighting from sensory drive we scaled only the recurrent weight contribution:

$$\begin{aligned} z_t^{(4)} &= W_{\text{in}}x_t + b_{\text{in}} + gW_{\text{rec}}h_{t-1} + b_{\text{rec}}, \\ h_t &= \tanh\big[(1-\alpha)h_{t-1} + \alpha z_t^{(4)}\big]. \end{aligned} \tag{8}$$

Recurrent bias remained outside the gain. P4 is a mechanistic contrast within the trained RNN, not a direct REBUS or 5-HT2A claim.

P5 Gaussian state noise

Independent Gaussian noise is retained in the project catalogue as a generic disruption control against which selective signatures must later be shown to exceed non-specific damage. Noise was added to the pre-activation,

$$h_t = \tanh[(1-\alpha)h_{t-1} + \alpha z_t + \sigma\varepsilon_t], \quad \varepsilon_t \sim \mathcal{N}(0, I), \tag{9}$$

in the exploratory pilot, but P5 was outside the candidate-only screen reported in Section . Section is reserved for the matched-cost comparison.

P6 State persistence

Trained working-memory RNNs can store information in slow or persistent latent dynamics [1, 2]. P6 asks whether weakening retention of the carried state, without rescaling incoming drive, can produce selective behavioural costs. The update is asymmetric:

$$h_t = \tanh[p(1-\alpha)h_{t-1} + \alpha z_t], \tag{10}$$

where p is the persistence scale. This is an operational RNN intervention, not a pre-specified biological mapping from psilocybin to reduced persistence; $p = 0.95$ entered the present screen as the exploratory pilot nominee.

P7 Conserved effective time constant

To test whether P6 effects are merely those of any timescale change, P7 rescales the leaky-integrator coefficients while preserving their sum. With timescale factor s,

$$\alpha' = \min(\alpha/s, 1), \qquad h_t = \tanh[(1-\alpha')h_{t-1} + \alpha' z_t]. \tag{11}$$

Thus $s > 1$ slows the effective timescale and $s < 1$ speeds it. Within the evaluated grid, $\alpha/s < 1$, so the minimum was inactive. P7 is a mechanistic control for conserved timescale change, not a psychedelic parameter claim.

Evaluation and behavioural signature

Each circular checkpoint, operator, strength, condition, and delay used 1,024 trials. Each N-back checkpoint, operator, strength, and load used 1,024 20-item sequences. Native and perturbed evaluations reused identical generated task samples within checkpoint and cell.

Circular response accuracy was mean absolute angular error. Post-cue settling was the first response step at which angular error was below 15° and population-vector amplitude exceeded 50% of the native reference amplitude. Trials that did not settle were capped at the 25-step response duration, yielding restricted mean settling time. Settling was interpreted only when fixation accuracy was at least 0.90 and at least 80% of trials settled. The 15° settling threshold is a coarse latency criterion distinct from continuous accuracy; half-native amplitude excludes collapsed bumps; and the 80% settled floor marks latency invalid when most trials never settle. “Slowing with preservation” required a positive settling change at clean delay 20 while proportional angular-error impairment remained at or below 0.20, encoding selective slowing without large accuracy loss.

For circular condition c, proportional impairment was

$$I_c = \frac{E_{\text{perturbed},c} - E_{\text{native},c}}{E_{\text{native},c}}, \tag{12}$$

where E is mean angular error. Delay selectivity was $I_{\text{delay }80} - I_{\text{delay }10}$; distractor selectivity was $I_{\text{distractor }20} - I_{\text{clean }20}$.

N-back discriminability was $D = H - F$, where H and F are hit and false-alarm rates. Condition-normalised impairment was

$$J_c = \frac{D_{\text{native},c} - D_{\text{perturbed},c}}{D_{\text{native},c}}, \tag{13}$$

and load selectivity was $J_{\text{2back}} - J_{\text{0back}}$. Positive values indicate disproportionate high-load impairment.

The candidate-screen analysis was descriptive. For each operator-strength setting we reported the checkpoint mean, SD, and a student t 95% interval across seeds, together with the fraction of seeds in the predicted direction. A setting met the complete descriptive pattern when the mean slowing-with-preservation, distractor, and N-back load contrasts were positive and at

Table II. Candidate perturbations and fixed evaluation grids. Neutral settings reproduce the native forward pass. Distractor-window gain was defined only for the circular distractor condition; all other operators except P5 were evaluated in both task families. P5 is listed for catalogue continuity but was not part of the candidate-only screen.

ID	Computational intervention	Strengths
P1	Scalar gain on input and recurrent weight drive	0.90–1.10
P2	Fixed mean-one unit-wise log-normal gain	log-SD 0.00–0.30
P3a	Gain on stimulus-channel input contribution	0.80–1.20
P3b	P3a restricted to irrelevant cue window	1.00–1.50
P4	Scalar gain on recurrent weight contribution	0.90–1.10
P5	Additive pre-activation noise (catalogue)	specificity chapter
P6	Scale p on carried-state coefficient	0.90–1.10
P7	Conserved update with $\alpha' = \min(\alpha/s, 1)$	0.80–1.25

least a simple majority of checkpoints agreed on each (three of five circular; six of ten N-back); delay selectivity was supporting evidence only. That majority rule generalises the exploratory pilot's $\geq$2-of-3 criterion and is implemented in the summary code; it was not restated as a numeric threshold in the 1,024-trial preregistration text. Cross-operator comparisons are uncontrolled for overall perturbation magnitude, so a near-miss may fail because of dose as well as mechanism. No participant-level inference, confirmatory p values, or multiple-comparison claims were made for the screen.

RESULTS: BASELINE COMPETENCE AND HIDDEN-STATE GEOMETRY

Before comparing perturbations, the retained circular and N-back checkpoints must be competent on their native tasks and must exhibit interpretable hidden-state organisation. This section reports those baselines. Pool competence uses the frozen held-out banks already recorded at screening; delay-sweep and dynamics panels use representative retained seeds (circular seed 20260735; N-back seed 20260916).

Distractor-trained circular baselines

Across the five retained distractor-trained circular checkpoints, held-out mean angular error averaged 2.70° on clean delay-20 trials and 3.54° with a mid-delay distractor (Fig. 2). All retained seeds passed the preconfigured competence gates (clean $\leq 10°$, distractor $\leq 15°$, fixation accuracy $\geq$ 0.90). Seed 20260734 failed those gates and was excluded before the perturbation screen; it is shaded in the pool figure for transparency.

On the representative retained seed, clean mean error rose only gradually with delay (1.71°, 1.86°, 2.10°, and 2.61° at delays 10, 20, 40, and 80), while the distractor delay-20 cell remained low at 2.92° (Fig. 3). Hidden-state PCA showed angle-ordered trajectories occupying a ring-like plane (Fig. 4A; PC1+PC2 = 70.4% variance). Step-to-step hidden-state speed slowed during the silent delay (Fig. 4B), and a ridge decoder fit on delay-period hidden states recovered cue angle with mean delay error 0.41° (Fig. 4C). Because the fixation-gated population readout is trained to remain silent until the go cue, that decoder is a hidden-state diagnostic rather than the behavioural output itself.

Competence-screened N-back baselines

All ten screened N-back checkpoints passed untouched competence gates (Fig. 5). 0-back accuracy was at ceiling in every seed; 2-back accuracy ranged from 0.958 to 0.986, with discriminability (hit rate minus false-alarm rate) from 0.917 to 0.971 and 2-back lure accuracy from 0.964 to 0.996. On representative seed 20260916, the frozen evaluation bank gave 0-back accuracy/discriminability of 1.000 and 2-back values of 0.986/0.971 (Fig. 6).

Hidden-state geometry differed by rule (Fig. 7A). At stimulus samples, 0-back activity formed tight clusters by current stimulus identity, whereas 2-back geometry was more overlapping, consistent with an updating demand that cannot be solved from the current item alone. Hidden-state speed under 2-back remained phasic across the sequence (Fig. 7B). Match probability for scored 2-back items separated match from non-match events within each item window (Fig. 7C).

These baselines establish that both families solve their native tasks with interpretable internal structure before any literature-motivated perturbation is applied.

RESULTS I: CANDIDATE PERTURBATION SCREEN

This section reports the completed candidate-versus-native-baseline screen. It nominates a leading operator but does not yet establish specificity against generic damage.

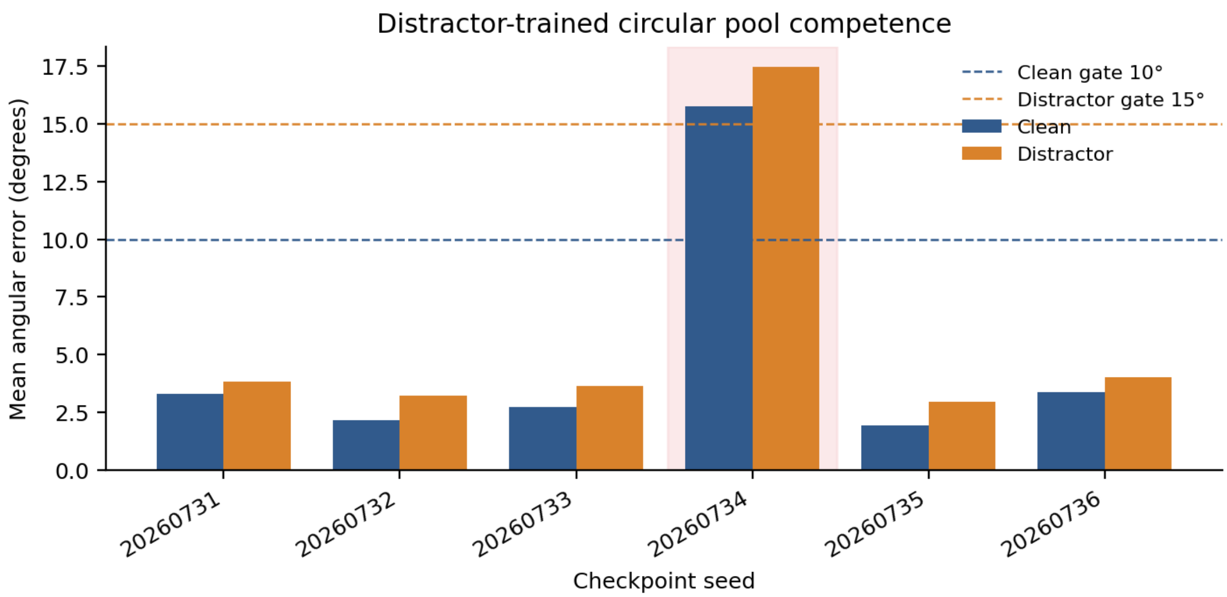

Figure 2. Clean versus distractor mean angular error for every trained circular seed. Retained checkpoints lie well below the competence gates; the failed seed is shaded.

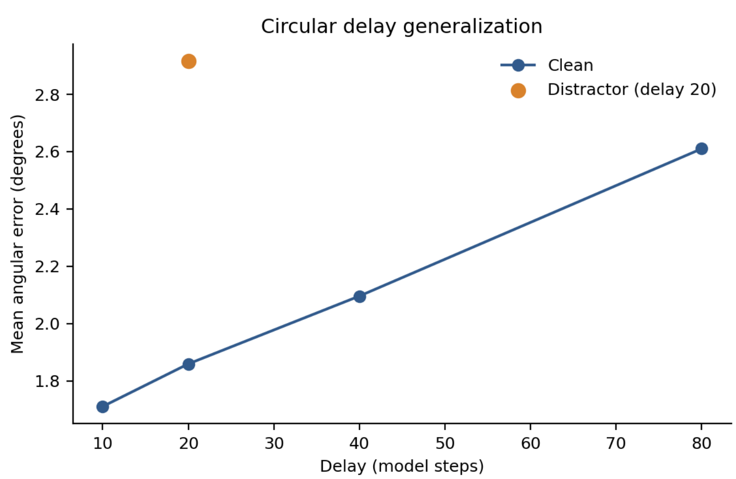

Figure 3. Clean delay generalization for representative circular seed 20260735, with the distractor delay-20 cell marked separately.

Nomination and independent circular retest isolated one complete profile

State persistence 0.95 was first nominated by an exploratory three-seed pilot on an earlier circular family without trained distractors. The present screen retested all candidate strengths on the new distractor-trained circular family and reused the completed ten-checkpoint N-back grid. Across the 23 non-neutral shared operator-strength profiles, only persistence 0.95 again met the complete descriptive majority pattern (Fig. 8). It showed slowing with preservation, long-delay selectivity, and distractor selectivity in four of five circular checkpoints, together with load selectivity in all ten N-back checkpoints. Relative to the pilot, circular delay selectivity shrank from 0.290 to 0.089 and distractor selectivity from 0.148 to 0.015, while N-back load selectivity remained similar (0.296 to 0.250). Neutral settings returned zero contrast. Other perturbations often reproduced one or two components, but not the full combination.

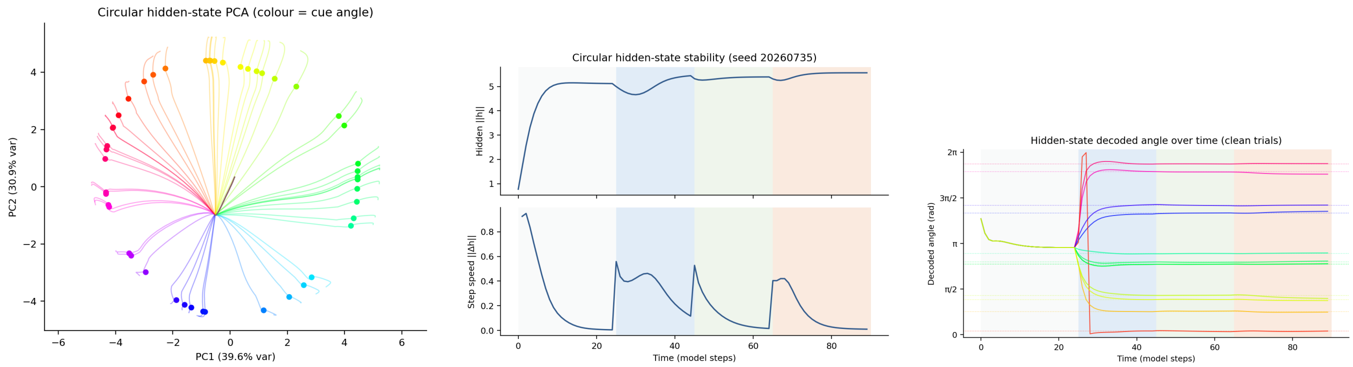

Figure 4. Circular hidden-state organisation on seed 20260735. **A**, PCA trajectories coloured by cue angle. **B**, Hidden-state norm and step speed across trial phases. **C**, Hidden-state decoded angle over time on clean trials.

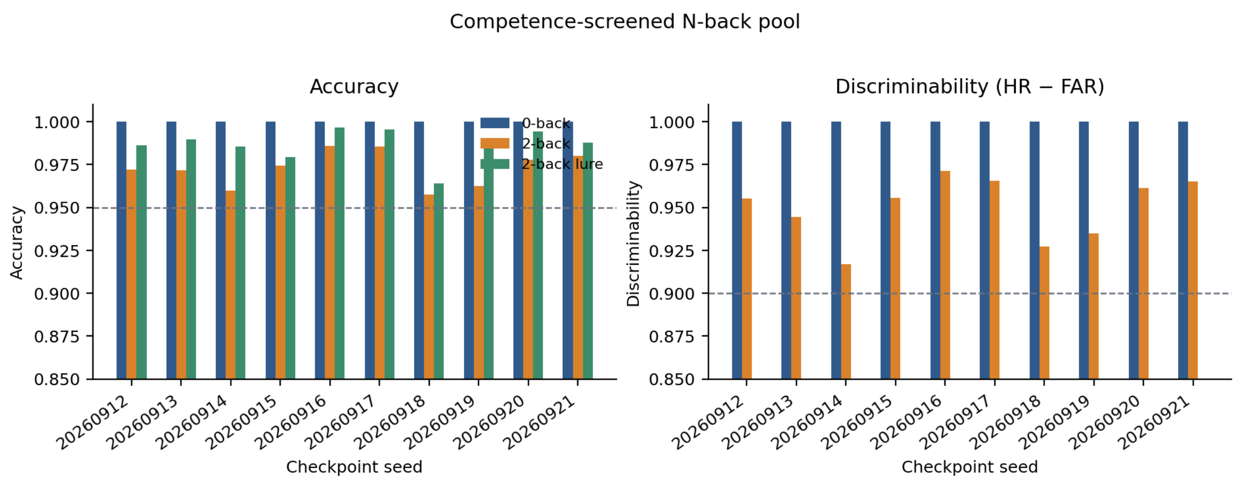

Figure 5. 0-back versus 2-back accuracy, discriminability, and 2-back lure accuracy across the ten retained N-back checkpoints.

A 5% persistence reduction preserved ordinary accuracy while selectively affecting demanding conditions

At $p = 0.95$, clean delay-20 proportional angular-error change was 0.049 ± 0.092 across circular checkpoints (95% t-interval $[-0.065, 0.164]$) and remained below 0.20 in $5/5$ seeds. Restricted mean settling increased by 0.104 ± 0.167 steps ($[-0.103, 0.312]$) and was positive in $4/5$ seeds, with a range of -0.013 to 0.393 steps (Table III). The circular settling, delay, and distractor intervals all include zero, so those components are directional majority tendencies rather than robust across-checkpoint effects. By contrast, N-back load selectivity was 0.250 ± 0.077 ($[0.195, 0.305]$), positive in $10/10$ checkpoints, and ranged from 0.110 to 0.363. Long-minus-short delay selectivity was 0.089 ± 0.075 ($[-0.004, 0.181]$) and distractor-minus-clean selectivity was 0.015 ± 0.044 ($[-0.040, 0.071]$), each positive in $4/5$ checkpoints. Thus the leading perturbation retained the predicted average ordering, but only the N-back contrast clearly excluded zero under the seed-level t-interval.

The persistence effect was dose-like but not monotonic across all features

Reducing persistence further to 0.90 enlarged mean settling, delay, and N-back load effects (Fig. 9). However, slowing with preservation passed only $1/5$ circular checkpoints, so the stronger intervention no longer expressed the selective accuracy-preserving pattern. Increasing persistence above one continued to impair N-back preferentially but reversed the circular delay and distractor contrasts. The joint signature was therefore specific to a weak reduction around 0.95 rather than monotonic damage at any non-neutral strength.

Alternative operators reproduced partial signatures

Sensory-input gain 1.20 produced distractor selectivity in $5/5$ circular checkpoints and load selectivity in $9/10$

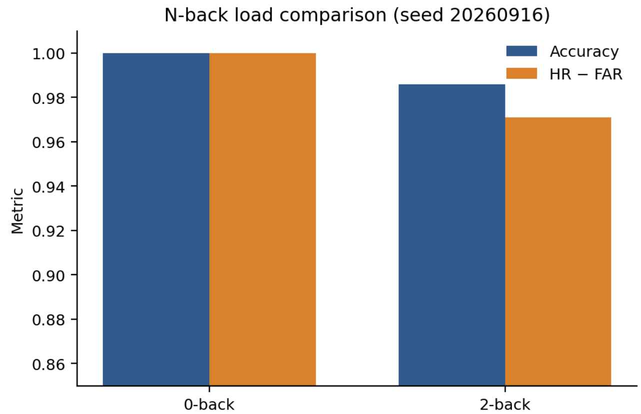

Figure 6. Representative N-back load comparison on seed 20260916 under the frozen evaluation bank.

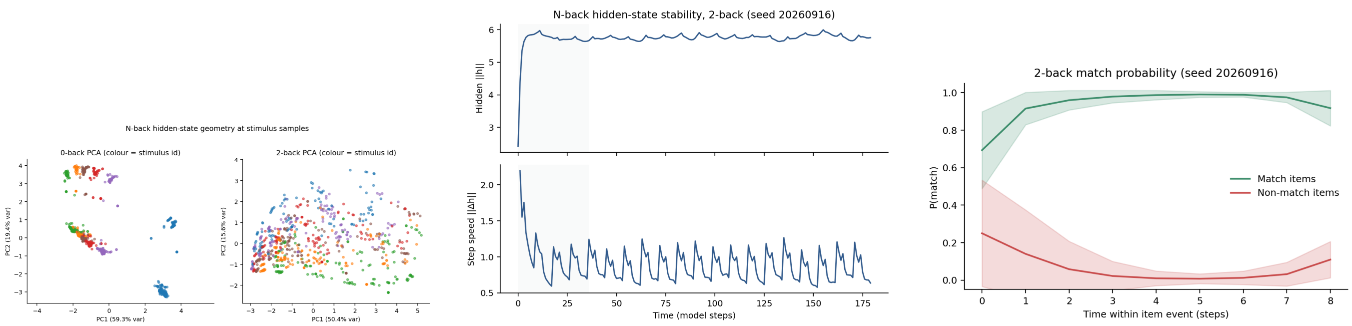

Figure 7. N-back hidden-state organisation on seed 20260916. **A**, PCA at stimulus samples coloured by stimulus identity under 0-back and 2-back rules. **B**, Hidden-state norm and step speed during 2-back sequences. **C**, Mean match probability for match versus non-match scored items.

N-back checkpoints, but mean settling became slightly faster (-0.010 ± 0.023 steps; positive in only $1/5$) and delay selectivity was positive in only $1/5$. Effective time-constant scale 1.10 slowed settling with preservation in $5/5$ (0.605 ± 0.176 steps) and showed load selectivity in $10/10$, but distractor selectivity was negative in every circular checkpoint. Conversely, time-constant scale 0.90 showed delay selectivity in $5/5$ and distractor selectivity in $4/5$ but accelerated settling in every circular checkpoint (-0.681 ± 0.170 steps). Synaptic-drive gain 1.05 showed delay selectivity in $4/5$, distractor selectivity in $3/5$, and load selectivity in $10/10$, but also accelerated settling in $5/5$ (-0.496 ± 0.240 steps). All of these near-miss clean delay-20 settling cells were latency-valid; uniqueness was therefore not an artefact of invalid latency scoring. These near-misses constrain the result: matching one difficult condition was insufficient to qualify as the complete pattern, and the discrimination against the conserved time-constant operator rested on settling direction and distractor selectivity rather than on large absolute settling magnitude.

The contrast between persistence and the conserved time-constant operator is mechanistically informative. Both reduced persistence ($p < 1$) and reduced timescale ($s < 1$) can weaken retention, but only persistence leaves the drive coefficient unchanged (Eq. 10). Reducing the conserved timescale ($s = 0.90$) accelerated post-cue settling while preserving delay and distractor selectivity; increasing it ($s = 1.10$) slowed settling but reversed distractor selectivity. The joint pattern therefore depended on the asymmetric carried-state intervention, not merely on any reduction in effective memory lifetime. Cross-operator comparisons remain uncontrolled for overall clean-task cost, so dose differences could still contribute to some near-misses.

Trained distractor filtering resolved the prior validity failure

All 945 circular result rows passed the fixation gate, all ten N-back native checkpoints passed competence, and all $p = 0.95$ clean delay-20 cells were latency-valid. Twelve rows elsewhere in the circular grid were latency-invalid (all heterogeneous-drive settings with low settled fraction); none of the near-miss settling comparisons above used those cells. Native distractor mean error now ranged from 3.46° to 4.83°, and every native distractor

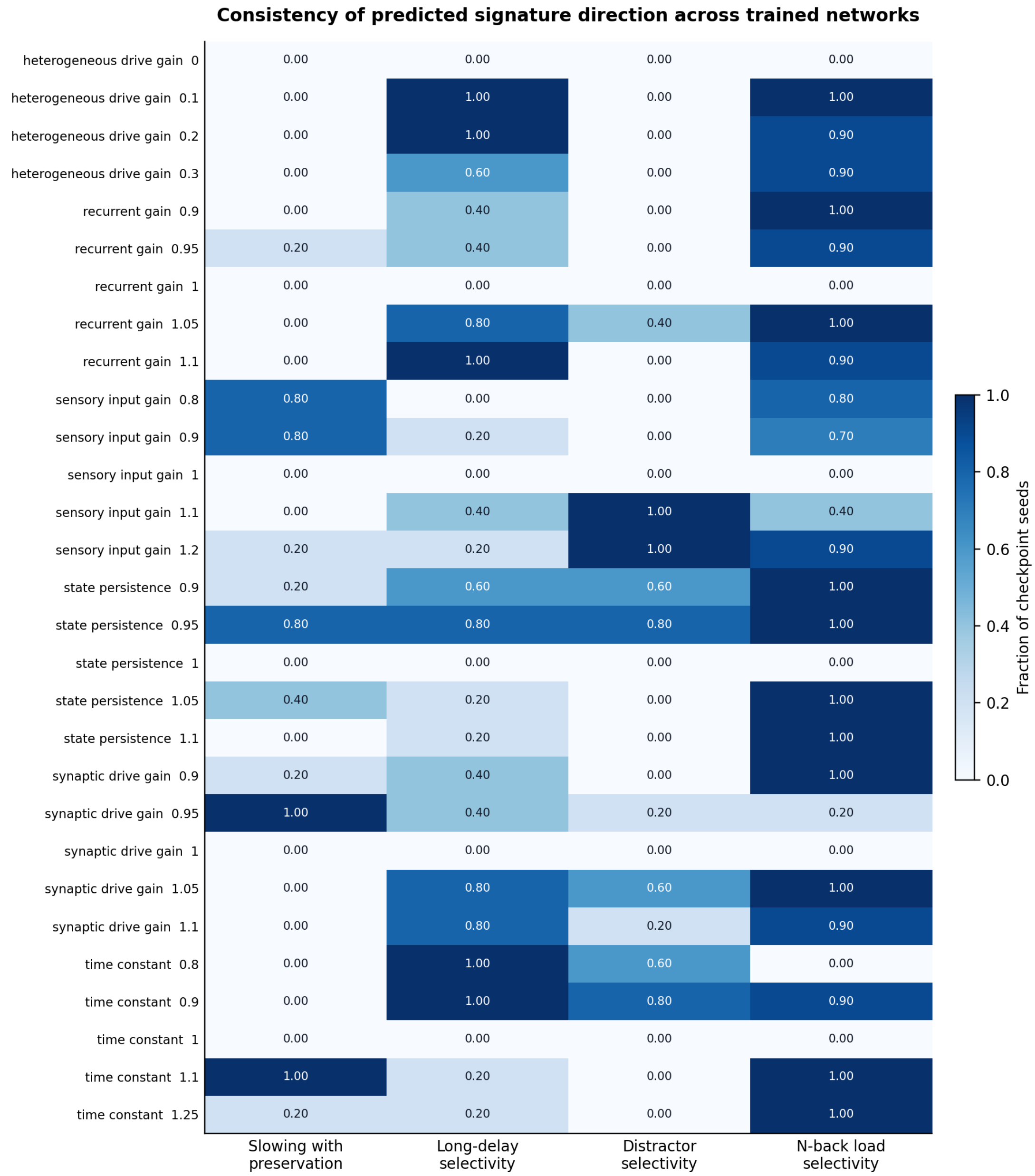

Figure 8. Cross-task signature screen. Rows are every shared operator-strength setting evaluated in both task families; columns show the fraction of independent checkpoint seeds with the predicted signature direction. Circular fractions have denominator five and N-back fractions denominator ten. State persistence 0.95 was the only row to meet the complete majority pattern and additionally reached 4/5 agreement on each circular feature with complete N-back agreement. Values quantify directional consistency, not participant-level probability or statistical significance.

cell had a settled fraction of 1.00. The distractor contrast therefore measured perturbation of a learned filtering behaviour rather than failure on an unfamiliar task. As a manipulation check, distractor-window gains 1.10, 1.25, and 1.50 increased distractor error by 3.7%, 9.8%, and 21.5% on average, respectively, with positive effects in 5/5 checkpoints. However, persistence-related distractor selectivity remained small and was negative in one checkpoint, so the improved validity does not convert the result into a strong human effect-size match.

A three-seed exploratory pilot had already suggested that persistence 0.95 can exceed weak Gaussian state noise on N-back load selectivity at nearly matched 0-back cost (means 0.296 versus 0.052 at noise SD 0.025). That observation is labelled secondary, used only three N-back checkpoints, and did not yield an informative circular cost match within the pilot's 0.05 tolerance; it is therefore not a specificity claim for the present screen.

Table III. State persistence 0.95. Values are mean ± SD across five circular or ten N-back checkpoint seeds; intervals are student t 95% intervals across those seeds.

Outcome	Contrast	95% interval	Consistency
Delay-20 error	0.049 ± 0.092	$[-0.065, 0.164]$	below 0.20 in 5/5
Settling change (steps)	0.104 ± 0.167	$[-0.103, 0.312]$	4/5 positive
Delay selectivity	0.089 ± 0.075	$[-0.004, 0.181]$	4/5 positive
Distractor selectivity	0.015 ± 0.044	$[-0.040, 0.071]$	4/5 positive
N-back load selectivity	0.250 ± 0.077	$[0.195, 0.305]$	10/10 positive

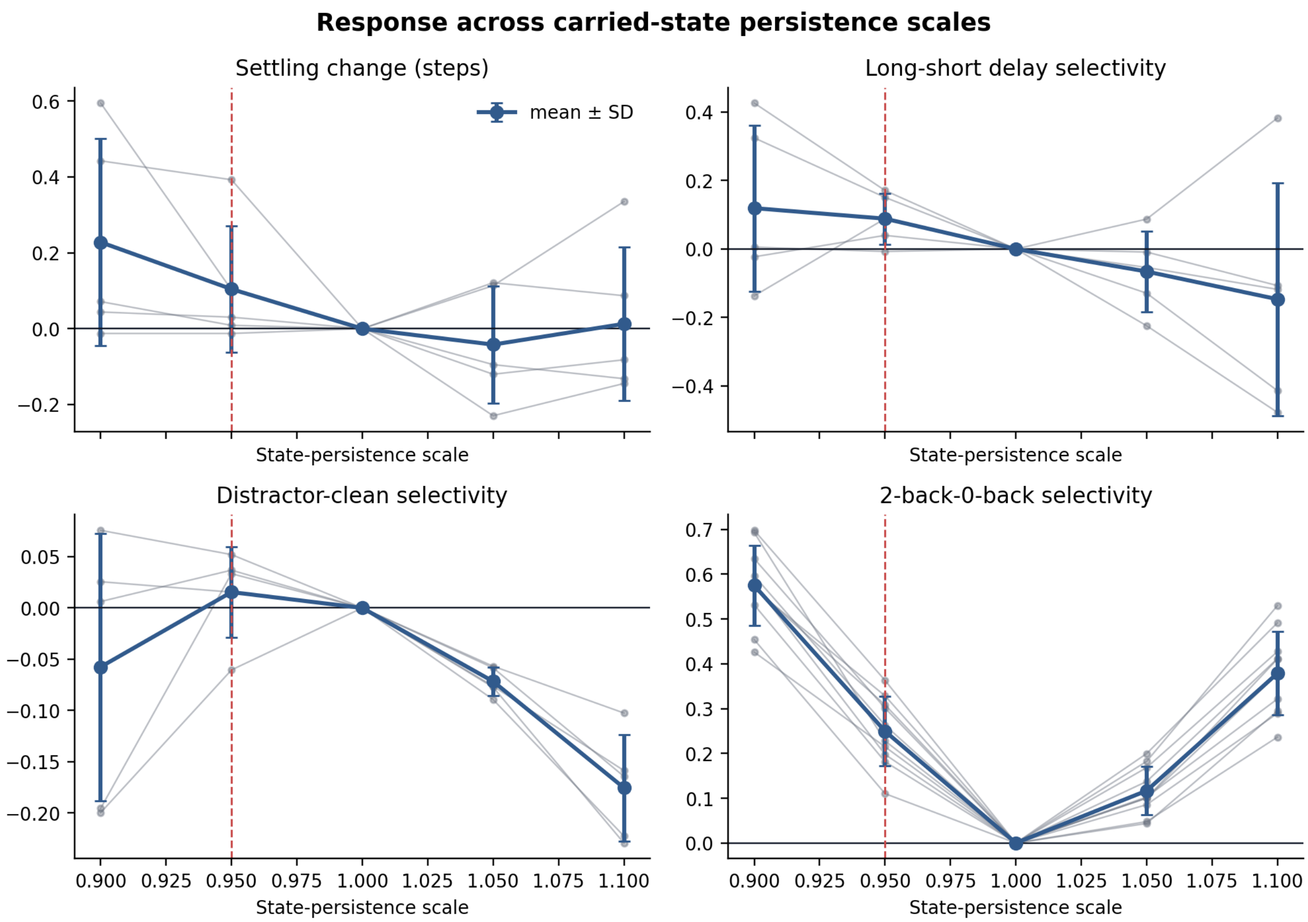

Figure 9. Dose-like response to carried-state persistence scaling. Grey lines are independently trained checkpoint seeds; blue points and error bars are mean ± SD. The dashed red line marks the leading scale 0.95 and the horizontal line marks no change from native performance. A stronger reduction increased several impairments but also violated relative preservation, whereas increased persistence reversed circular delay and distractor selectivity.

RESULTS II: SPECIFICITY AGAINST GENERIC DISRUPTION

Scaffold / to write.
This section is required by the dissertation end goal and is not yet written. Intended content:

- Pre-register persistence 0.95 versus matched-cost Gaussian state noise (P5) on the trained-distractor circular family and the competence-screened N-back family.
- Match on clean-task proportional cost before comparing excess settling, delay, distractor, and load selectivity.
- Report whether the selective pattern survives after cost matching.

- Keep circular and N-back outcomes separate; do not pool raw scores.

Until this section exists, the dissertation can claim native-baseline sufficiency for persistence 0.95, but not comparative mechanism against generic damage.

RESULTS III: DYNAMICAL EXPLANATION

Scaffold / to write.

This section is required by the dissertation end goal and is not yet written. Intended analyses for the retained candidate (and matched controls):

- delay-period decoding and representational drift;
- distractor-induced displacement and recovery;
- memory-subspace / population geometry;
- late-delay state entropy or related diversity measures;
- interaction with learned recurrent connectivity where interpretable.

The goal is to explain *why* any selective behavioural pattern emerges, not only that it appears in behavioural metrics.

DISCUSSION

Present interpretation of the candidate screen

The nomination-and-retest screen identified reduced carried-state persistence as the strongest computational account of the selected cross-task behavioural pattern against the native baseline. Scaling the carried-state coefficient by 0.95 left ordinary circular accuracy approximately intact while modestly slowing response settling, increasing vulnerability at longer delays and under distraction, and selectively impairing 2-back discriminability.

The evidential weight is uneven across tasks. The N-back load contrast was robust: mean selectivity 0.250 ± 0.077 with a seed-level 95% interval that excluded zero and agreement in all ten checkpoints. By contrast, the circular settling, delay, and distractor contrasts had the predicted mean signs and majority directional agreement (4/5), but their across-checkpoint intervals all included zero. In plain terms, the dissertation currently has one solid behavioural pillar—disproportionate high-load updating impairment—and three weaker circular hints. The "complete pattern" label therefore reflects the prespecified majority-direction rule, not equal robustness of every component.

Equation 10 still supplies a coherent model-level reading. Each update retains slightly less of the preceding hidden state without simultaneously rescaling incoming drive. Information need not be erased at once; its stability can weaken cumulatively, so short and simple conditions remain relatively intact while longer maintenance, competing input, and repeated updating expose the loss. The conserved time-constant near-misses support the asymmetry interpretation: when carried-state and drive coefficients are rescaled together, the joint signature breaks. The present behavioural analyses do not yet demonstrate a changed attractor or slow manifold; that is reserved for Section .

Relation to the human target profile

The selective human findings reviewed in Section were used as ordinal targets, not quantitative fits. The robust N-back result aligns in direction with Barrett-style load dependence [3]. The circular slowing-with-preservation motif is only a directional analogue of Vollenweider/Yousefi-style RT dominance [6, 7]: settling is externally cued, the mean change is far below one model step, and the seed-level interval includes zero. Distractor selectivity is likewise only a weak directional analogue of Carter-style filtering vulnerability [4]; the circular task tests a related principle, not multiple-object tracking, and the effect size shrank sharply once distractor filtering became a trained competence rather than an out-of-distribution probe. Delay selectivity is also weak after retest, despite cleaner validity. Thus the human-facing claim must currently be stated as: persistence 0.95 reproduces the selected load dissociation convincingly and the remaining circular dissociations only as small majority tendencies.

Claim boundary

"State persistence" is an operational property of this implemented recurrence, not a measured neurobiological parameter. The current result shows computational sufficiency against the native baseline for the descriptive majority pattern, carried primarily by N-back load selectivity. It does not show that psilocybin reduces neural persistence, that 5-HT2A activation maps to $p = 0.95$, or that the mechanism is unique. A matched generic-damage comparator remains the outstanding specificity test (Section); the exploratory Gaussian observation is only a provisional three-seed N-back hint. Several alternative operators also reproduced individual components, and cross-operator comparisons remain uncontrolled for overall clean-task cost.

Limitations of the present evidence

Several limitations constrain inference now. First, post-cue settling is not literal reaction time because response onset is externally imposed. Second, circular effect sizes are small and checkpoint-heterogeneous; one checkpoint reversed the distractor contrast and another slightly reversed delay selectivity. Third, circular delay selectivity shrank several-fold from the pilot family to the distractor-trained retest on clean trials alone, so effect size is family-dependent. Fourth, the descriptive screen used many operator-strength profiles without a confirmatory inferential family. Fifth, checkpoint consistency demonstrates robustness across trained models, not population inference about human participants. Sixth, N-back reuse of three pilot seeds means that arm is only partly independent of nomination.

Next empirical steps

The immediate next experiment should compare persistence 0.95 with matched-cost Gaussian disruption on the present circular and N-back families (Section). If specificity fails on N-back, the leading candidate should be demoted regardless of the descriptive screen. If specificity holds on N-back but circular contrasts remain weak, the dissertation story should be rebalanced around load selectivity as the primary result, with circular outcomes as secondary directional support. Only after that comparison should persistence be densified around 0.95 and delay-period representational drift quantified (Section).

CONCLUSION

A 5% reduction in carried-state persistence was the only tested RNN perturbation to reproduce the complete descriptive majority pattern after exploratory nomination and retest on distractor-trained circular networks. The strongest present evidence is selective 2-back impairment across all ten N-back checkpoints. Circular slowing, delay, and distractor contrasts had the predicted average direction in four of five checkpoints, but their effects were small and their seed-level intervals included zero. Training with distractors removed the previous out-of-distribution confound without converting those circular contrasts into robust findings.

Reduced persistence therefore remains a credible candidate for matched-cost specificity and dynamical analysis, but the present result is an abstract descriptive sufficiency finding against the native baseline—carried mainly by the load dissociation—not evidence for a biological mechanism of psilocybin.

Scaffold / to write.

Rewrite after Sections and are complete. Target story:

> We built and validated working-memory RNNs, established how generic variability disrupts them, derived candidate dynamical perturbations from psychedelic theory and human task evidence, and tested which, if any, reproduces the selective behavioural and representational signature better than nonspecific noise.

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